



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
School of Engineering and Engineering Technology

Department of Electrical and Electronics Engineering
B. Eng. 1st Semester 2023/2024 Examination

EEE 505 – Digital Signal Processing

Date: 21/05/2024

Instruction: Answer any three questions Time: 3 Hrs

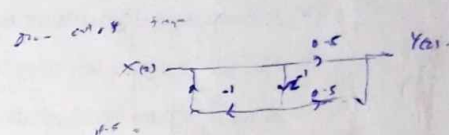
QUESTION 1

- (a) Design a single pole low pass digital filter with a 3 dB bandwidth (3 dB cut off frequency), $\omega_c = 0.5\pi$ by applying bilinear transformation to the analog Butterworth filter

$H(s) = \frac{\Omega_c}{s + \Omega_c}$ where Ω_c is the cut off frequency of analog filter

Determine;

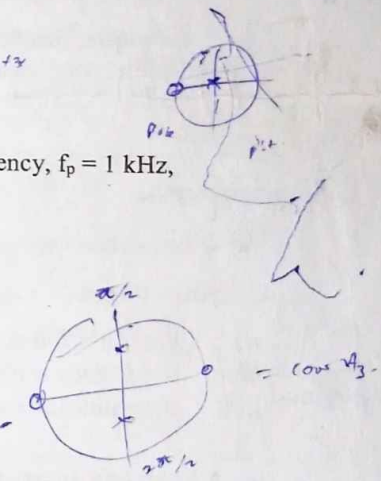
- The system transfer function, $H(z)$,
 - The difference equation, and
 - The coefficient of the filter.
 - Hence, Implement the filter with suitable structure
- (b) (i) Plot the unit circle for the system developed in Q1a
(ii) Determine the stability status of system developed in Q1a



QUESTION 2

- (a) An FIR low pass filter is required to have pass band edge frequency, $f_p = 1$ kHz,
Transition band, Δf 200 Hz
Pass band ripple, $A_p = 0.1$ dB
Minimum stop band attenuation $A_c = 50$ dB
Sampling frequency, $F_s = 10$ kHz

- Choose an appropriate window with a reason for the design
 - Obtain four coefficients of the filter
 - Implement with suitable structure
 - Determine the impulse response of the filter
 - What is the response of the filter to a unit sample with $\delta(n)$
- (b) Determine the frequency response with odd filter length and positive symmetrical impulse response



QUESTION 3

- (a) Use the Figure Q3a to determine,
(i) Transfer function, $H(z)$ of the system given that the pole is at radius, R and

(ii) The difference equation

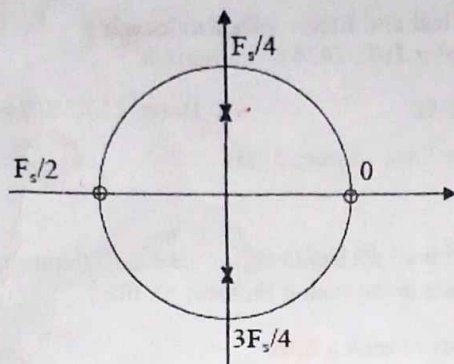


Figure Q3a

(b) A band pass digital filter is required to meet the following specifications

Complete signal rejection at DC and 500 Hz

Transition width 10 Hz.

A narrow pass band centred at 250 Hz

Assuming a sampling frequency of 1 kHz obtain,

- the transfer function of the filter, $H(z)$, and
- the filter coefficient and difference equation
- Hence, implement the filter.

QUESTION 4

(a) Given a causal system

$$y(n) = 0.9y(n-1) + x(n)$$

- Find $H(z)$ and sketch the pole-zero plot. Hence, determine the stability status
- Determine the $h(n)$
- Determine the response of the system at DC, $\frac{\pi}{2}$ and π

(b) A linear time-invariant system is determined by the difference equation

$$y(n) - y(n-1) + 0.25 y(n-2) = x(n) + 2x(n-1) + x(n-3)$$

- Determine the system stability with reason
- What is the System response to a unit step sequence

QUESTION 5

- a. An analog signal $x(t)$ is described as

$$x(t) = 4 + 2\cos(150\omega t) + 4\cos(350\omega t)$$

The signal is sampled at 200 Hz, obtain the discrete time signal, $x(n)$ and comment on your result with reason(s). Suggest possible solution to address identified problem.

- (b) An analog signal, $x(t)$ is sampled to obtain discrete time signal, $x(n)$

$$x(t) = 1 + \sin 2\pi ft$$

where f is 1 kHz and F_s is 4 kHz

Obtain:

- The sequence, $x(n)$; and
- The Amplitude spectra of the signal
- Plot the spectra

Filter type	$h_D[n], n \neq 0$	$h_D[n], n = 0$
Low-pass	$2F_c \frac{\sin(n\Omega_c)}{n\Omega_c}$	$2F_c$

Name of window function $w(n)$	Mathematical definition
Rectangular	1
Hanning	$0.5 - 0.5 \cos \left[\frac{2\pi n}{N-1} \right]$
Hamming	$0.54 - 0.46 \cos \left[\frac{2\pi n}{N-1} \right]$
Blackman	$0.42 - 0.5 \cos \left[\frac{2\pi n}{N-1} \right] + 0.08 \cos \left[\frac{4\pi n}{N-1} \right]$

Name of window function $w[n]$	Transition width ΔF in (Hz), (normalised)	Pass-band ripple A_p in (dB)	Ripple δ_p, δ_s	Side-lobe level in (dB)	Stop-band attenuation A_s in (dB)
Rectangular	$0.9/N$	0.741	0.089	-13	21
Hanning	$3.1/N$	0.0546	0.063	-31	44
Hamming	$3.3/N$	0.0194	0.0022	-41	53
Blackman	$5.5/N$	0.0017	0.000196	-57	74



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B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: EEE505 **COURSE TITLE:** Digital Signal Processing

UNITS: 3

TIME ALLOWED: 3 HOURS

Instruction: ANSWER FOUR Questions in all.

Question 1

- (a) Show that signal to quantization noise ratio in dB is expressed as $SNR = 10 \log P_s + 10.8 + 6n - 20 \log D$, where n is the number of bits, D the dynamic range of the signal and P_s is the signal power.
- (ii) Consider a sinusoid $x(t) = A \sin \omega t$, show that SNR (dB) is linear with n as the slope.
- (b)(i) Consider a continuous time ramp function $x(t) = 4t$ over a time interval of 0 to 1 sampled at 0.1s using a 3-bit quantizer. Compute the sampled signal, the quantized signal, the quantization error and the signal to quantization noise ratio in dB
- (c) Obtain SNR using the derived formula for the ramp signal and compare your result to the statistical value in 1(c)(ii) (note P_s is for ramp signal)

Question 2

- (a)(i) Compute and plot the amplitude spectral of $x(n) = (1, 2, 1, 0)$ using FFT algorithm. Discuss the plot.
- (b) Obtain the original signal from its spectral using matrixes method.
- (c) Consider a continuous signal $x(t)$ sampled at (i) 40 Hz and (ii) 160 Hz. Discuss the results $x(t) = 3 \cos 20\pi t + 4 \cos 40\pi t + 10 \cos 80\pi t$

Question 3

- (a)(i) Given a LTI system, $y(n) = 0.9y(n-1) + x(n)$. (i) Find the system function. (ii) the pole zero diagram (iii) the impulse response (iv) the step response.
- (ii) Given that an LTI system is described by the following difference equation $y(n) = 0.8y(n-2) + x(n) - x(n-2)$
- (b)(i) Derive the system function
- (ii) determine the stability status with reason(s)
- (iii) compute the frequency response at 0 and $\pi/2$

Question 4

- (a) Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired filter are $0.9 \leq H(\omega) \leq 1$; $0 \leq \omega \leq \frac{\pi}{2}$; $H(\omega) \leq 0.2$; $\frac{3\pi}{4} \leq \omega \leq \pi$ with $T=1s$,

$$H(s) = \frac{\Omega_c}{s + \Omega_c} \times \frac{\Omega_c^2}{s^2 + s\Omega_c b k + \Omega_c^2}$$

- (i) Obtain the order N of the analog filter, the cutoff frequency and $H(s)$. (ii) the $H(z)$ and the coefficients of the filter (iii) Realize the filter with a suitable structure.
- (b) Design a bandpass filter using pole-zero placement method with the following specifications: center frequency 100 Hz; passband 10 Hz; Stopband edge frequencies 50 Hz and 100 Hz; and. Sampling



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Frequency 400 Hz.

(i) Obtain the system function (ii) sketch the unit circle and determine its stability status (iii) the difference equation

Question 5

- (a) Derive the impulse response of type 1 FIR filter and show that the phase is linear
- (b) A FIR filter is required to have the following specifications passband edge frequency 2 kHz; Stopband edge frequency 2.2 kHz; passband ripple 0.016; Stopband ripple 0.00316 and sampling Frequency 10 kHz.
- (i) Select appropriate window function with a reason (ii) calculate the required numbers of coefficients and the group delay (iii) determine the filter impulse response between $n=0$ and $n=4$. (iv) Realise the filter with a suitable structure

Window Type

Rectangular Hanning

Hamming

Blackman

Mathematical function $1, 0.5-0.5\cos 2\pi n/(N-1), 0.54 - 0.46\cos 2\pi n/(N-1), 0.42 - 0.5\cos 2\pi n/(N-1) + 0.08\cos 2\pi n/(N-1)$

Transition width $0.9/N, 3.1/N, 3.3/N, 5.5/N$

Stopband attenuation 21 44 53 74

Question 6

- (a) A LTI system function $H(Z) = x(z) + 2z^{-1}x(z) + 3z^{-2}x(z) + 4z^{-3}x(z) + 5z^{-4}x(z) + 6z^{-5}x(z) + 7z^{-6}x(z) + 8z^{-7}x(z)$. What is the response of the system to a sinusoidal input $x(n) = \cos 0.25\pi n$
- (b) Consider an analog filter described by $H(s) = \frac{n_c}{s + n_c}$. Design a digital filter whose magnitude at cut off frequency is 3 kHz equals the magnitude of $H(s)$ at analog frequency of 2 rad/sec if the sampling rate is 12 kHz using bilinear transformation by obtaining the difference equation and filter coefficients.
- (c) Show with equation how to recover a signal buried in noise.